C++ Programming

1. Summary

History and Purpose

Bjarne Stroustrup created C++ in 1980 at Bell Laboratories. He built it on top of C, adding object-oriented programming (OOP) without changing the core of C. Because of this, C and C++ share most of their syntax, but C++ adds classes and objects, which C does not have.

Why C++ Is Used

- It is one of the most popular languages in the world.
- It powers operating systems, GUIs, and embedded systems.
- OOP gives programs clear structure and reusable code.
- It is portable across many platforms.

C vs C++

C is procedural and works top-down through functions. C++ is multi-paradigm: it supports procedural, object-oriented, and generic programming, and can be built top-down or bottom-up. Only C++ supports classes, objects, inheritance, polymorphism, encapsulation, function overloading, and try/catch exception handling. C manages memory manually with malloc/free; C++ uses new/delete plus constructors and destructors. C uses printf/scanf; C++ uses cout/cin.

Data Types

C++ data types fall into four groups: basic types (int, char, float, double, void), derived types (array, pointer, reference), enumeration types (enum), and user-defined types (struct, class, union, typedef). Common sizes: char = 1 byte, short/int = 2 bytes, long int = 4 bytes, float = 4 bytes, double = 8 bytes, long double = 10 bytes.

Operators

C++ has five main operator groups:

- Arithmetic: + - * / % (multiplication and division run before addition and subtraction)
- Relational: == != > < >= <= (compare two values, return true/false)
- Bitwise: & | ^ ~ << >> (work on individual bits)
- Logical: && || ! (combine or reverse true/false conditions)
- Assignment: = += -= *= /= %= (store or update a value in a variable)

Identifiers, Variables, and Constants

An identifier is the name given to a variable, function, or other program item. It must start with a letter or underscore (never a digit), can contain letters, digits, and underscores, is case-sensitive, and cannot be a reserved keyword. A variable is a named memory location whose value can change while the program runs. A constant is declared with the const keyword and its value cannot change once set.

Basic Program Structure

A C++ program has three parts: the standard library section (#include lines), the main function (int main()), and the function body, where the actual code runs. Every program needs #include<iostream> and using namespace std; to use cout and cin.

Control Statements

Control statements decide the order in which code runs. They fall into three groups: decision-making (if, if-else, if-else-if, switch), looping (for, while, do-while), and jump statements (break, continue). The for loop is best when you know how many times to repeat. The while loop checks its condition before running. The do-while loop always runs its body at least once because it checks the condition after. A switch statement compares one value against several cases and needs break to stop it from running the next case too.

Arrays

An array stores many values of the same data type under one name, in memory locations next to each other. C++ arrays are index-based, starting at 0. For example, int A[10] can hold 10 integers, from A[0] to A[9]. You can loop through an array with a for loop to read or update every element.

Functions

A function is a named group of statements that performs one task. It can take parameters (data passed in) or none at all. Functions make programs shorter and easier to reuse, since you write the logic once and call it whenever needed.

Classes and Objects

A class is a blueprint that groups variables (state) and functions (behaviour) together. An object is a real instance created from that class. Every object has state (its data), behaviour (what it can do), and identity (its unique name). Classes support encapsulation (bundling data and methods, restricting access), inheritance (one class reusing another's features), and polymorphism (the same function behaving differently depending on context).

2. Practice Questions

A. True or False (20 Questions)

1. C++ was developed by Bjarne Stroustrup in 1980.
2. C++ does not support object-oriented programming.
3. C supports classes and objects, just like C++.
4. C++ is a multi-paradigm language.
5. C++ uses new and delete for memory management.
6. C uses cout and cin for input and output.
7. C++ supports inheritance.
8. A char data type uses 1 byte of memory.
9. A double data type uses 8 bytes of memory.
10. The % operator gives the remainder of a division.
11. The == operator is used to assign a value.
12. Bitwise operators work on individual bits.
13. The && operator is the logical OR operator.
14. An identifier can start with a digit.
15. C++ identifiers are case-sensitive.
16. A constant's value can change while the program runs.
17. An if-else-if ladder can test more than one condition.
18. A break statement stops a switch case from falling into the next case.
19. The first index of a C++ array is 1.
20. An object is an instance of a class.

True/False Answers: 1-True, 2-False, 3-False, 4-True, 5-True, 6-False, 7-True, 8-True, 9-True, 10-True, 11-False, 12-True, 13-False, 14-False, 15-True, 16-False, 17-True, 18-True, 19-False, 20-True

B. Multiple Choice (20 Questions)

1. Who developed the C++ language?
a) Dennis Ritchie
b) Bjarne Stroustrup
c) James Gosling
d) Guido van Rossum

2. In which year was C++ developed?
a) 1970
b) 1975
c) 1980
d) 1990

3. Which of these does C NOT support?
a) Functions
b) Loops
c) Classes
d) Arrays

4. Which header file is used for input/output in C++?

a) stdio.h
b) iostream
c) conio.h
d) math.h

5. Which operator is used for multiplication?

a) /
b) %
c) *
d) +

6. What is the result of 10 % 3?

a) 3
b) 1
c) 0
d) 10

7. Which operator checks if two values are equal?

a) =
b) ==
c) !=
d) <=

8. Which of these is a bitwise operator?

a) &&
b) ||
c) &
d) !

9. Which operator reverses a true/false value?

a) &&
b) ||
c) !
d) ==

10. Which of these is a valid identifier?

a) 2data
b) break
c) _sum
d) Sum-1

11. Which keyword is used to declare a constant in C++?

a) var
b) const
c) static
d) fixed

12. What is the memory size of a float?

a) 2 byte
b) 4 byte
c) 8 byte
d) 10 byte

13. Which loop always runs its body at least once?

a) for
b) while

c) do-while
d) switch

14. Which statement chooses code to run based on matching case values?
a) if
b) while
c) switch
d) for

15. Which keyword exits a loop early?
a) continue
b) break
c) return
d) exit

16. What is the index of the first element in a C++ array?
a) 1
b) -1
c) 0
d) None

17. Which keyword is used to define a class in C++?
a) struct
b) object
c) class
d) new

18. An object is best described as:
a) A blueprint
b) An instance of a class
c) A function
d) A loop

19. Which operator combines addition and assignment?
a) =
b) +=
c) ==
d) --

20. Which C++ feature provides containers like vector and map?
a) stdio.h
b) iostream
c) Standard Library (STL)
d) math.h

MCQ Answers: 1-b, 2-c, 3-c, 4-b, 5-c, 6-b, 7-b, 8-c, 9-c, 10-c, 11-b, 12-b, 13-c, 14-c, 15-b, 16-c, 17-c, 18-b, 19-b, 20-c

C. Matching Items (20 Questions)

Match each term in Column A with its correct description in Column B.

Column A (Term)	**Column B (Description)**
1. Encapsulation	A. A memory location whose value cannot change during the program
2. Polymorphism	B. A class gets properties and behaviour from another class
3. Inheritance	C. Compares two values, e.g. ==, >, <
4. Class	D. Combines data and methods, and restricts access to them
5. Object	E. Assigns or updates a value, e.g. =, +=
6. Array	F. An instance of a class
7. Function	G. A symbol that performs an operation on data, e.g. +, -
8. Constant	H. A name given to a variable, function, or other program item
9. Variable	I. Combines two or more conditions, e.g. &&, ||
10. Identifier	J. The same function behaves differently depending on context
11. do-while Loop	K. A collection of same-type elements stored together in memory
12. Operator	L. Runs one block of code if true, another if false
13. Switch statement	M. Helps avoid naming conflicts in a program
14. if-else statement	N. A blueprint used to create objects
15. Arithmetic operator	O. A memory location whose value can change during the program
16. Relational operator	P. Works on individual bits of a number
17. Logical operator	Q. A group of statements grouped to perform one task
18. Bitwise operator	R. Runs its body once before checking the condition
19. Assignment operator	S. Works on numbers, e.g. +, -, *, /
20. Namespace	T. Picks one block of code to run from several case values

Matching Answers: 1-D, 2-J, 3-B, 4-N, 5-F, 6-K, 7-Q, 8-A, 9-O, 10-H, 11-R, 12-G, 13-T, 14-L, 15-S, 16-C, 17-I, 18-P, 19-E, 20-M

D. Fill in the Blanks (20 Questions)

1. C++ was created by ______ at Bell Laboratories.

2. C++ supports ______ programming, which C does not support.

3. The ______ operator divides two numbers and returns the remainder.

4. Variables and constants are both examples of C++ ______.

5. The keyword used to declare a constant in C++ is ______.

6. A ______ is the name of a memory location whose value can change.

7. The ______ loop checks its condition only after running the body once.

8. The ______ statement picks a block of code to run by matching case values.

9. Arrays in C++ start indexing from ______.

10. A ______ is a blueprint used to create objects.

11. An ______ is an instance of a class.

12. The ______ operator (!) reverses a true/false value.

13. ______ lets a class inherit properties from another class.

14. The header file used for input and output in C++ is ______.

15. cout is used for ______ and cin is used for ______.

16. A double data type uses ______ bytes of memory.

17. ______ means the same function behaves differently depending on context.

18. The ______ statement stops a loop before it finishes normally.

19. C++ uses ______ and ______ for dynamic memory, instead of malloc and free.

20. ______ helps avoid name conflicts in large programs.

Fill in the Blanks Answers: 1-Bjarne Stroustrup, 2-object-oriented, 3-% (modulus), 4-identifiers, 5-const, 6-variable, 7-do-while, 8-switch, 9-0, 10-class, 11-object, 12-logical NOT, 13-Inheritance, 14-iostream, 15-output / input, 16-8, 17-Polymorphism, 18-break, 19-new / delete, 20-Namespace

E. Practical Code Questions (4 Questions)

1. Write a C++ program that reads an integer and prints whether it is even or odd.
2. Write a C++ program that reads three numbers and prints the largest one using if-else statements.
3. Write a C++ program that reads a number and prints its multiplication table (1 to 10) using a for loop.
4. Write a C++ program that stores 5 integers in an array and prints their total sum using a for loop.

F. Explanation / Mixed Code Questions (4 Questions)

1. Look at this code:

```
int a = 10, b = 3;
cout << a % b;
What does it print, and why?
```

2. Explain what happens if a switch statement is missing the break statement after each case. Use a short example to show the effect.

3. Look at this code:

```
for (int i = 1; i <= 3; i++) {
  cout << i * i << " ";
}
What is the output, and why?
```

4. Explain the main difference between a while loop and a do-while loop. Give one short code example for each.